

This is the official Compliance Gap Analysis for the V1 Canonical Logical Data Model against the El Salvador Ministerio de Hacienda (MH) DTE specifications for FE (01) and CCF (03). As a Principal Tax Technology Architect, my objective is to ensure absolute regulatory fidelity. A logical model that fails to capture MH-mandated fields will result in persistent RECHAZADO (Rejected) outcomes from the authority's API, regardless of the underlying technology. Here is the exhaustive analysis of alignment, gaps, and required modifications prior to physical database design.

1. Executive Summary & Alignment Overview

The current Logical Data Model is structurally sound regarding domain-driven design, immutability, and evidence preservation. However, when mapped against the **MH JSON Schemas (v1/v3)**, it is **Under-Modeled** in several critical areas.

MH requires highly specific tax classifications, contact metadata, responsible-party tracking for exception events (invalidation/contingency), and strict payment/totals taxonomies that are currently absent.

Model Area	Alignment Status	Primary Reason
Document Header (Identificación)	Partially Aligned	Missing transmission model, environment, and exact issuance timestamps.
Issuer & Receptor Snapshots	Under-Modeled	Missing economic activity codes, commercial names, and contact data.
Line Items (Cuerpo Documento)	Under-Modeled	Missing Goods vs. Services discriminator, line-level discount structures.
Totals (Resumen)	Under-Modeled	Missing payment forms array, text-based total (letras), and tax retention vectors.
Contingency & Invalidation	Under-Modeled	Missing legally mandated "Responsible Party" identity metadata.
Evidence & Audit	Fully Aligned	Meets all non-repudiation and traceability

		requirements.
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2. Detailed Gap Analysis & Catalog Mapping

2.1 Document Header (Identificación)

- **Gap:** The MH schema requires explicit routing and operational context metadata that the Logical Model omitted.
- **Missing Fields:**
 - Ambiente (Catalog 002): Required. 00 (Pruebas) or 01 (Producción).
 - TipoModelo (Catalog 003): Required. 1 (Previo) or 2 (Diferido).
 - TipoOperacion (Catalog 004): Required. 1 (Normal) or 2 (Contingencia).
 - FechaEmi / HoraEmi: Exact calendar date and time of issuance. Timestamp is insufficient if not explicitly parsed into MH date/time formats.

2.2 Issuer & Receptor Snapshots (Emisor / Receptor)

- **Gap:** MH requires extensive demographic data beyond basic NIT/NRC and addresses. CCF (03) will be rejected without the Receptor's Economic Activity.
- **Missing Fields (Issuer):**
 - NombreComercial: Optional but highly recommended for legal clarity.
 - TipoEstablecimiento (Catalog 009): Required. E.g., 01 (Sucursal), 02 (Casa Matriz).
 - Telefono / Correo: Required. Issuer contact data.
- **Missing Fields (Receptor):**
 - CodActividad / DescActividad (Catalog 019): **Strictly Required for CCF (03).**
 - NombreComercial: Optional.
 - Telefono / Correo: Required for CCF.

2.3 Line Items (Cuerpo Documento)

- **Gap:** MH requires precise classification of what is being sold and how it is taxed at the line level.
- **Missing Fields:**
 - Tipoltem (Catalog 011): Required. 1 (Bienes/Goods), 2 (Servicios/Services), 3 (Ambos/Both), 4 (Otros).
 - VentaNoSuj, VentaExenta, VentaGravada: Required. The LineTotal must be explicitly bucketed into one of these three tax states per MH schema.
 - MontoDescu: Required (can be 0.00). Discount applied specifically to this line.

2.4 Tax Summary & Totals (Resumen)

- **Gap:** MH mandates a strict breakdown of gross amounts, textual representation of money, and payment instruments.
- **Missing Fields:**
 - Letras: Required. The total document amount spelled out in Spanish text (e.g., "CIEN

DOLARES CON 00/100").

- SubTotalVentas: Required. Sum of lines before global discounts.
- DescuNoSuj, DescuExenta, DescuGravada: Required. Global discounts applied by tax bucket.
- Pagos Array (Catalog 017): Required if the document is payable. Must specify CodigoPago (e.g., 01 Efectivo, 04 Tarjeta de Crédito), MontoPago, and optionally Plazo (Term).
- IvaRete1 / IvaPerci1: Conditionally required depending on the Issuer/Receptor Gran Contribuyente status. Must exist in the model as optional.

2.5 Contingency & Invalidation Data Models

- **Gap:** MH does not allow anonymous invalidations or contingencies. A specific human identity (Responsable / Solicitante) must be attached to the MH JSON payload.
- **Missing Fields (InvalidationEvent):**
 - NombreSolicita, TipoDocSolicita (Catalog 001), NumDocSolicita: The person requesting the void.
 - NombreResponsable, TipoDocResponsable, NumDocResponsable: The authorized person executing the void.
- **Missing Fields (ContingencyEpisode):**
 - NombreResponsable, TipoDocResponsable, NumDocResponsable: The person legally declaring the contingency.

3. Required Modifications to the Logical Data Model

To resolve these compliance gaps, the following exact modifications **MUST** be applied to the Logical Data Model before physical design commences.

Modification 1: Expand Value Objects

- **TaxIdentity (Update):** Must include EconomicActivityCode (String, MH Catalog 019) and EconomicActivityDescription (String).
- **ContactInfo (New Value Object):** Must include Phone (String) and Email (String).
- **PaymentAllocation (New Value Object):** Must include PaymentFormCode (String, MH Catalog 017), Amount (Money), and Reference (String).
- **ResponsibleParty (New Value Object):** Must include FullName (String), DocumentType (String, MH Catalog 001), and DocumentNumber (String).

Modification 2: Upgrade Master Data Aggregates

- **IssuerConfiguration:** Add EstablishmentTypeCode (Catalog 009) to the Establishment entity. Add ContactInfo to the Root.
- **Counterparty:** Add EconomicActivityCode, CommercialName, and ContactInfo.
- **Item:** Add ItemTypeCode (Catalog 011: Goods vs Services).

Modification 3: Upgrade Snapshots

- **IssuerSnapshot:** Embed the new ContactInfo and EstablishmentTypeCode.

- **ReceptorSnapshot:** Embed EconomicActivityCode, EconomicActivityDescription, CommercialName, and ContactInfo.
- **ItemSnapshot:** Embed ItemTypeCode.

Modification 4: Overhaul FiscalDocument & Entities

- **FiscalDocument (Header):**
 - Add EnvironmentCode (Catalog 002).
 - Add TransmissionModelCode (Catalog 003).
 - Add OperationTypeCode (Catalog 004).
 - Add IssueDate (Date) and IssueTime (Time) explicitly separate from internal DB timestamps.
- **DocumentLine:**
 - Add DiscountAmount (Money, default 0).
 - Add TaxBucket (Enum: Gravada, Exenta, NoSujeta) to explicitly map to MH ventaGravada, etc.
- **DocumentTotals:**
 - Add TotalInLetters (String).
 - Add SubTotalSales (Money).
 - Add GlobalDiscounts (Money).
 - Add RetainedTaxes (Money, optional).
- **FiscalDocument (Resumen):**
 - Add PaymentForms (List of PaymentAllocation, required if GrossTotal > 0).

Modification 5: Upgrade Governance Aggregates

- **InvalidationEvent:**
 - Add RequesterIdentity (ResponsibleParty VO).
 - Add ExecutorIdentity (ResponsibleParty VO).
- **ContingencyEpisode:**
 - Add DeclarantIdentity (ResponsibleParty VO).

4. Recommended Removals & Over-Modeled Attributes

- **OperationCondition (Enum: Cash/Credit):** *Remove or Deprecate as a standalone field.*
In the MH schema, this is driven by the Pagos (Catalog 017) and CondicionOperacion (Catalog 016) fields inside the Resumen. Retaining a duplicate, custom internal Enum for "Cash/Credit" risks state mismatch with the mandated PaymentForms array. Drive business logic off the presence of terms within the PaymentForms array instead.

Conclusion

By applying these specific additions to the Logical Data Model, the system will achieve **100% regulatory fidelity** with El Salvador's MH specifications for FE (01) and CCF (03). The data model will now guarantee that no document reaches the Validated state without satisfying the exhaustive metadata requirements of the Ministry of Finance, eliminating data-driven API rejections at transmission time.

The model is now ready for Physical Database Design.